FOR A FLAT MODULE THE PRODUCT OF TWO IDEALS APPLIED TO M MEETS ITSELF AS THE PRODUCT OF THE INTERSECTION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The exact sequence 0 to I cap J to R to R/I plus R/J. Written down at the first line of the counted proof (line 8867). Presents the intersection of the two ideals as the kernel of a map out of R.
- Flatness as exactness of tensoring. Applied at proof lines 8868-8871. Preserves the exact sequence after tensoring with M, so that the intersection is carried into (I cap J) tensor M.

Lemma 0.1. *Let R be a ring. Let $I, J \subset R$ be ideals. Let M be a flat R-module. Then $IM \cap JM = (I \cap J)M$.*

Proof. Consider the exact sequence $0 \to I \cap J \to R \to R/I \oplus R/J$. Tensoring with the flat module M we obtain an exact sequence

$$0 \to (I \cap J) \otimes_R M \to M \to M/IM \oplus M/JM$$

Since the kernel of $M \to M/IM \oplus M/JM$ is equal to $IM \cap JM$ we conclude. $\square$

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